

Zhangir Nurmukhambetov

+31 06 38965908 | zhangir.dev@gmail.com | jon-gear.github.io

EDUCATION

Breda University of Applied Sciences

Bachelor of Science in Creative Media and Game Technologies

September 2023 – Expected Jun 2026

Breda, Netherlands

TECHNICAL SKILLS

Languages: C++, C#, GLSL, HLSL, Python

Frameworks & Libraries: OpenGL, EnTT (ECS), ImGui, Cereal, GLM, tinygtf, ImGuizmo

Game Engines: Unreal Engine 5 (Blueprints & C++), Unity (C#), Godot

Developer Tools: Visual Studio, Visual Studio Code, Git, Perforce (P4V), GitHub Actions

Languages: English (Fluent), Russian (Fluent), Arabic (Beginner)

Interests: Creative writing, Digital art and design

SELECTED PROJECTS

Zentera | *Custom C++ Engine, Angelscript, FMOD*

May 2025 – Jun 2025

- Led as Programming Rep, improving sprint communication and reducing inter-team delays by 40%.
- Built a Unity-style **Particle System** using EnTT ECS, cutting frame drops by 60%.
- Integrated 20+ FMOD **sound events** (e.g., footsteps, tongue, environment) for immersive feedback.
- Resolved 30+ gameplay bugs and led the Itch.io release, including trailer content and performance specs.
- Facilitated cross-disciplinary collaboration, aligning visual/audio features with design goals to enhance player experience.

Fire Ant Engine | *C++, OpenGL, GLSL, EnTT, ImGui, Cereal*

Feb 2025 – Apr 2025

- Built a real-time terrain editor with brush tools and prefab support—used across 100% of game maps.
- Implemented GPU-based terrain editing and tessellation rendering, reducing latency by 90%.
- Created JSON-based scene serialization with EnTT Meta, halving iteration time on level testing.
- Optimized logic and rendering to maintain stable FPS on large terrains and multiple platforms.
- Contributed to flow-field navigation and pathfinding systems, enabling smooth AI movement across complex terrain.

Custom Terrain Editor | *C++, OpenGL, GLSL, ImGui, GLM*

Nov 2024 – Jan 2025

- Developed a GPU-accelerated terrain sculpting tool with modular brush system and intuitive UI.
- Implemented heightmap deformation with dynamic normals and 80%+ reduction in visual latency.
- Designed foundational architecture reused in Fire Ant's terrain editor, accelerating team ramp-up and integration.

Custom C++ Game Engine | *C++, EnTT, ImGui, Cereal, STB, tinygtf, ImGuizmo*

Sep 2024 – Nov 2024

- Built modular ECS engine with runtime editor, particle systems, scene serialization, and GLTF support.
- Enabled real-time editing, transform gizmos, and reusable asset management for future projects.
- Implemented platform abstractions for input, rendering, and file I/O—supporting PC and console builds.

IgKnighthed | *Unreal Engine 5*

May 2024 – Jun 2024

- Developed 7+ gameplay systems (e.g., input, health, AI, power-ups) for a twin-stick bullet-hell shooter.
- Implemented couch co-op with full controller support, enhancing accessibility and showcase playtesting.
- Contributed to successful itch.io launch with 250+ plays and positive feedback on gameplay feel.
- Used Blueprints to rapidly iterate across design, animation, and combat systems.
- Balanced enemy behavior and damage systems, ensuring fair difficulty scaling in local multiplayer.